

The power of differentiation

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Topics

- The big idea: optimisation by following gradients
- Recap: what are gradients and how do we find them?
- Recap: Singular Value Decomposition and its applications
- Example: Computing SVD using gradients - The Netflix Challenge

The big idea: optimisation by following gradients

- Fundamentally, we're interested in machines that we train by optimising parameters
 - How do we select those parameters?
- In deep learning/differentiable programming we typically define an objective function that we *minimise* (or *maximise*) with respect to those parameters
- This implies that we're looking for points at which the gradient of the objective function is zero w.r.t the parameters

The big idea: optimisation by following gradients

- Gradient based optimisation is a *big* field!
 - First order methods, second order methods, subgradient methods...
- With deep learning we're primarily interested in first-order methods¹.
 - Primarily using variants of gradient descent: a function $F(\mathbf{x})$ has a minima² (or a saddle-point) at a point $\mathbf{x} = \mathbf{a}$ where $\mathbf{a}$ is given by applying $\mathbf{a}_{n+1} = \mathbf{a}_n - \alpha \nabla F(\mathbf{a}_n)$ until convergence from some initial point $\mathbf{a}_0$.

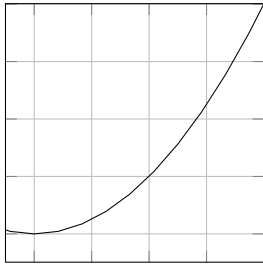
Recap: what are gradients and how do we find them?

The derivative in 1D

- Recall that the gradient of a straight line is $\frac{\Delta y}{\Delta x}$.
- For an arbitrary real-valued function, $f(a)$, we can approximate the derivative, $f'(a)$ using the gradient of the *secant line* defined by $(a, f(a))$ and a point a small distance, h , away $(a + h, f(a + h))$: $f'(a) \approx \frac{f(a+h)-f(a)}{h}$.
 - This expression is 'Newton's Quotient' or 'Fermat's Difference Quotient'.
 - As h becomes smaller, the approximated derivative becomes more accurate.
 - If we take the limit as $h \rightarrow 0$, then we have an exact expression for the derivative: $\frac{df}{da} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$.

¹Second order gradient optimisers are potentially better, but for systems with many variables are currently impractical as they require computing the Hessian.

²not necessarily global or unique



Recap: what are gradients and how do we find them?

The derivative of $y = x^2$ from first principles

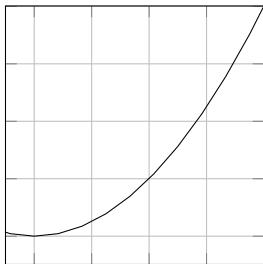
$$\begin{aligned}
 y &= x^2 \\
 \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2hx - x^2}{h} \\
 \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{h^2 + 2hx}{h} \\
 \frac{dy}{dx} &= \lim_{h \rightarrow 0} (h + 2x) \\
 \frac{dy}{dx} &= 2x
 \end{aligned}$$

Recap: what are gradients and how do we find them?

Aside: numerical approximation of the derivative

- For numerical computation of derivatives it is better to use a “centralised” definition of the derivative:

- $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h}$
- The bit inside the limit is known as the *symmetric difference quotient*
- For small values of h this has less error than the standard one-sided difference quotient.



Recap: what are gradients and how do we find them?

Aside: numerical approximation of the derivative

- If you are going to use difference quotients to estimate derivatives you need to be aware of potential rounding errors due to floating point representations.
 - Calculating derivatives this way using less than 64-bit precision is rarely going to be useful. (Numbers are not represented exactly, so even if h is represented exactly, $x + h$ will probably not be)
 - You need to pick an appropriate h - too small and the subtraction will have a large rounding error!

Recap: what are gradients and how do we find them?

Derivatives of deeper functions

- Deep learning is all about optimising deeper functions; functions that are compositions of other functions
 - e.g. $z = f \circ g(x) = f(g(x))$
- The chain rule of calculus tells us how to differentiate compositions of functions:
 - $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$

Recap: what are gradients and how do we find them?

Example: differentiating $z = x^4$

Note that this is a silly example that just serves to demonstrate the principle!

$$\begin{aligned} z &= x^4 \\ z &= (x^2)^2 = y^2 \quad \text{where } y = x^2 \\ \frac{dz}{dx} &= \frac{dz}{dy} \frac{dy}{dx} = (2y)(2x) = (2x^2)(2x) = 4x^3 \end{aligned}$$

Equivalently, from first principles:

$$\begin{aligned} z &= x^4 \\ \frac{dz}{dx} &= \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} \\ \frac{dz}{dx} &= \lim_{h \rightarrow 0} \frac{h^4 + 4h^3x + 6h^2x^2 + 4hx^3 + x^4 - x^4}{h} \\ \frac{dz}{dx} &= \lim_{h \rightarrow 0} h^3 + 4h^2x + 6hx^2 + 4x^3 = 4x^3 \end{aligned}$$

Recap: what are gradients and how do we find them?

Vector functions

- What if we're dealing with a *vector* function, $\mathbf{y}(t)$?
 - This can be split into its constituent coordinate functions: $\mathbf{y}(t) = (y_1(t), \dots, y_n(t))$.
 - Thus the derivative is a vector (the 'tangent vector'), $\mathbf{y}'(t) = (y_1'(t), \dots, y_n'(t))$, which consists of the derivatives of the coordinate functions.
 - Equivalently, $\mathbf{y}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{y}(t+h) - \mathbf{y}(t)}{h}$ if the limit exists.

Recap: what are gradients and how do we find them?

Functions of multiple variables: partial differentiation

- What if the function we're trying to deal with has multiple variables³ (e.g. $f(x, y) = x^2 + xy + y^2$)?
 - This expression has a pair of *partial derivatives*, $\frac{\partial f}{\partial x} = 2x + y$ and $\frac{\partial f}{\partial y} = x + 2y$, computed by differentiating with respect to each variable x and y whilst holding the other(s) constant.
- In general, the partial derivative of a function $f(x_1, \dots, x_n)$ at a point $(a_1, \dots, a_n)$ is given by: $\frac{\partial f}{\partial x_i}(a_1, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i+h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}$.
- The vector of partial derivatives of a scalar-value multivariate function, $f(x_1, \dots, x_n)$ at a point $(a_1, \dots, a_n)$, can be arranged into a vector: $\nabla f(a_1, \dots, a_n) = \left(\frac{\partial f}{\partial x_1}(a_1, \dots, a_n), \dots, \frac{\partial f}{\partial x_n}(a_1, \dots, a_n) \right)$.
 - This is the **gradient** of f at a .
- In the case of a vector-valued multivariate function, the partial derivatives form a matrix called the **Jacobian**.

³A multivariate function

Recap: what are gradients and how do we find them?

Functions of vectors and matrices: partial differentiation

- For the kinds of functions (and programs) that we'll look at *optimising* in this course have a number of typical properties:
 - They are scalar-valued
 - * We'll look at programs with *multiple losses*, but ultimately we can just consider optimising with respect to the *sum* of the losses.
 - They involve multiple variables, which are often wrapped up in the form of vectors or matrices, and more generally tensors.
 - **How will we find the gradients of these?**

Recap: what are gradients and how do we find them?

The chain rule for vectors

Suppose that $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$, g maps from $\mathbb{R}^m$ to $\mathbb{R}^n$ and f maps from $\mathbb{R}^n$ to $\mathbb{R}$.

If $\mathbf{y} = g(\mathbf{x})$ and $z = f(\mathbf{y})$, then

$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}.$$

Equivalently, in vector notation:

$$\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^T \nabla_{\mathbf{y}} z$$

where $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ is the $n \times m$ Jacobian matrix of g .

Recap: what are gradients and how do we find them?

The chain rule for Tensors

- Conceptually, the simplest way to think about gradients of tensors is to imagine flattening them into vectors, computing the vector-valued gradient and then reshaping the gradient back into a tensor.
 - In this way we're still just multiplying Jacobians by gradients.
- More formally, consider the gradient of a scalar z with respect to a tensor $\mathbf{X}$ to be denoted as $\nabla_{\mathbf{X}} z$.
 - Indices into $\mathbf{X}$ now have multiple coordinates, but we can generalise by using a single variable i to represent the complete tuple of indices.
 - * For all index tuples i , $(\nabla_{\mathbf{X}} z)_i$ gives $\frac{\partial z}{\partial \mathbf{x}_i}$.
 - Thus, if $\mathbf{Y} = g(\mathbf{X})$ and $z = f(\mathbf{Y})$ then $\nabla_{\mathbf{X}} z = \sum_j (\nabla_{\mathbf{X}} \mathbf{Y}_j) \frac{\partial z}{\partial \mathbf{Y}_j}$.

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Let $\mathbf{D} = \mathbf{X}\mathbf{W}$ where the rows of $\mathbf{X} \in \mathbb{R}^{n \times m}$ contain some fixed *features*, and $\mathbf{W} \in \mathbb{R}^{m \times h}$ is a matrix of weights.
- Also let $\mathcal{L} = f(\mathbf{D})$ be some scalar function of $\mathbf{D}$ that we wish to minimise.
- What are the derivatives of $\mathcal{L}$ with respect to the weights $\mathbf{W}$?

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Start by considering a specific weight, W_{uv} : $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}}$.
- We know that $\frac{\partial D_{ij}}{\partial W_{uv}} = 0$ if $j \neq v$ because D_{ij} is the dot product of row i of $\mathbf{X}$ and column j of $\mathbf{W}$.

- Therefore, we can simplify the summation to only consider cases where $j = v$: $\sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} \frac{\partial D_{iv}}{\partial W_{uv}}$.
- What is $\frac{\partial D_{iv}}{\partial W_{uv}}$?

$$\begin{aligned}
 D_{iv} &= \sum_{k=1}^m X_{ik} W_{kv} \\
 \frac{\partial D_{iv}}{\partial W_{uv}} &= \frac{\partial}{\partial W_{uv}} \sum_{k=1}^m X_{ik} W_{kv} = \sum_{k=1}^m \frac{\partial}{\partial W_{uv}} X_{ik} W_{kv} \\
 \therefore \frac{\partial D_{iv}}{\partial W_{uv}} &= X_{iu}
 \end{aligned}$$

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Putting every together, we have: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} X_{iu}$.
- As we're summing over multiplications of scalars, we can change the order: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i X_{iu} \frac{\partial \mathcal{L}}{\partial D_{iv}}$.
- and note that the sum over i is doing a dot product with row u and column v if we transpose X_{iu} to $X_{ui}^\top$:
 $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i X_{ui}^\top \frac{\partial \mathcal{L}}{\partial D_{iv}}$.
- We can then see that if we want this for all values of $\mathbf{W}$ it simply generalises to: $\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{D}}$.

Recap: what are gradients and how do we find them?

STOP! What does a gradient actually mean?

- In your early calculus lessons you likely had it hammered into you that gradients represent rates of change of functions.
- This is of course totally true...
- But, it isn't a particularly useful way to think about the gradients of a loss with respect to the weights of a parameterised function.
 - **The gradient of the loss with respect to a parameter tells you how much the loss will change with a small perturbation to that parameter.**